


# CDS6314 DATA MINING

## Trimester 1, 2024/2025

### PROJECT (30%)

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**INSTRUCTIONS:**

1. This project carries 30% of the coursework assessment.
2. This is a group project, with a maximum of 4 members.
3. Deliverables for this assignment include Python code (*.ipynb*), a report (*.pdf*) and presentation.
4. Timelines:
   - Proposal deadline: **Week 9, 24th May 2024 (Friday), 11.59pm**
   - Final submission deadline: **Week 13, 23rd June 2024 (Sunday), 11.59pm**.
   - Presentations schedule: **Week 14**
5. Late-Day policy applies (20% deduction per day late from deadline).
6. If ***plagiarism*** is detected, the assignment will be granted 0% with no negotiation.

**INTRODUCTION:**

In this project, your task is to perform data mining on structured data of your domain of interest. You will be required to devise the full pipeline, from dataset search till knowledge visualization to gain hands-on experience on the implementation of a data mining solution on practical data. This project will be split into 2 phases, the proposal phase, and the implementation phase.

**DATASET / TOPICS:**

1. Search for dataset(s) / topic in a domain of interest from online repositories to conduct this project.
2. Dataset(s) / topics can be something your group is interested in with the condition that the topic is not exactly the same as the FYP of any group member.
3. The selected dataset should be structured and have at least 20 attributes (columns) and 5,000 instances (rows).
4. Sources:
    - UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/index.php)
    - Kaggle (https://www.kaggle.com/data)
    - Malaysia's Open Data Portal (http://www.data.gov.my/data/en_US/dataset)
    - DOSM Malaysia (https://open.dosm.gov.my/data-catalogue)
    - World Bank Open Data (https://data.worldbank.org/)
    - The Humanitarian Data Exchange (https://data.humdata.org/)
    - Other relevant and legitimate sources.

**DESCRIPTION OF TASKS:**

**Phase 1 (Proposal) [Week 7 - 9]**.

Prepare a simple write-up of the selected dataset and preliminary details of the project.

The proposal should include the following details:

1. Cover page: Project members and project title
2. Project description:
    - *Dataset*:

      What is the domain of the dataset? What is the source? Briefly introduce the dataset (attributes, records, etc.) the collection process of the original source. (Optional: can include preliminary data exploration, statistics, etc.)
    - *Data Mining Task*:

      What is the data mining task you intend to execute? Explain the motivation, aims, and expected outcomes of the project.
    - *Related Works*:

      Which reading will you examine to provide context and background? Search for at least 4 related research papers within the last 5 years, either using the same / similar dataset.
    3. Preliminary Task Distribution
    4. Declaration of FYP
    5. References

**Phase 2 (Final) [Week 10 - 13].**

Final report and source code submission, and presentation.

Suggested structure for **Final Report**:

1. **Cover page:** Title, Authors
2. **Abstract:** Summary of overall project, from motivation to conclusion. (maximum 250 words)
3. **Introduction:** Introduce the background, motivations, and objectives.
4. **Literature review:**

   Review at least 4 related research papers that used the same / similar / related dataset.

   Discuss the similarities and differences of those papers with this project.

5. **Methodology**

   Describe the overall data mining pipeline implemented to achieve the project objectives and provide justification on the steps and methods selected. The report should include, but not limited to, the following items:

   a. *Overall framework*: introduce the overall pipeline and data mining task.
   b. *Dataset*: source, collection process, volume, attributes, etc.
   c. *Data Preprocessing*: EDA, feature selection, data transformation, etc.
   d. *Data Mining*: Techniques used, parameters, etc.
   e. *Evaluation*: Experiments conducted, evaluation metrics, comparisons, etc.
   f. *Results and Discussion*: Compile results generated (tables, charts, etc.) and discuss the outcomes and findings.
   g. (Optional) Deployment: Dashboard, web tool, etc. (Streamlit, Heroku, etc.)

6. **Conclusion**

   Summarize the overall findings of the work and discuss potential use case or importance. Suggest potential future directions of the work (e.g. how to overcome limitations, other dimensions of exploration, etc.)

7. **References:** APA format

8. **Appendix**

*Note: It is not necessary to screenshot and show the python codes in the technical report. Instead, use text descriptions, algorithms, visualizations/flowcharts, or others to explain the work.*

**Source Code** preparation:

Source code of application/implementation must be in Python.

Please ensure the code can be used in different machines and they can be in Python scripts (.py) or Python Notebook (.ipynb). If any special instructions are needed for building or running the code, please provide a readme file.

You may create separate python scripts/notebooks for each key steps, if necessary.

Please include as Appendix in the Final Report of any tutorials, GitHub codes, websites, videos, etc. used for learning and reference to complete the project.

**Presentation** guidelines:

Present the overall project from motivation to insights.

- All members of the group must attend the face-to-face presentation, and each should talk about one part of the work done.
- Every group must prepare slides for the presentation (max. 10 slides including title slide with group details and ending slide).
- Additional tools can be used to make the presentation more effective (figures, tables, animations, etc.).
- Prepare a short demonstration of the code during the presentation.
- Maximum duration of presentation: **15 minutes** including demo.

**SUBMISSION CHECKLIST:**

All submissions to be made via Ebwise. (Do NOT submit any zip file)

1. Phase 1: Proposal submission (.pdf).
   - Name the submission file following this format:

     Proposal_Group[Number].pdf (e.g. *Proposal_Group01.pdf*)
   - Deadline: **Week 9,** 24th **May 2024 (Friday), 11.59pm**
2. Phase 2: Final report (.pdf), Source code (.py / .ipynb), Presentation slides (.pptx), datasets (.csv)
   - Please ensure the codes can reproduce the results given the raw data.
   - Include a Readme.txt of instructions to navigate the dataset and codes (if necessary, especially if multiple data files and scripts)
   - Name your files appropriately:

     FinalReport_Group[Number].pdf (e.g. *FinalReport_Group01.pdf*)

     Notebook_Group[Number].ipynb (e.g. Notebook_Group01.ipynb)

     Presentation_Group[Number].pptx (e.g. Presentation_Group01.pptx)

     Dataset_Group[Number].csv (e.g. Dataset_Group01.csv)
   - Deadline: **Week 13, 23rd June 2024 (Sunday), 11.59pm**
3. Presentation:
   - Schedule: **Week 14**
   - Specific slots and location to be announced later.

**PROJECT RUBRICS:**

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|---|---|---|
| **Proposal** (5%) | | 5 |
| **Report** (8%) | Clarity, structure, language, reference format | 2 |
| | Understanding of the background and literature | 2 |
| | Motivations and objectives | 2 |
| | Data mining task formulation and justification | 2 |
| **Technical** (8%) | Correctness | 3 |
| | Depth | 3 |
| | Complexity | 2 |
| **Analysis** (4%) | Results comparison and analysis | 2 |
| | Discussion on findings and insights | 2 |
| Presentation (5%) | | 5 |
| | | |
| **Total** | | **30%** |